

# Mitigation of Misinformation (MoM)

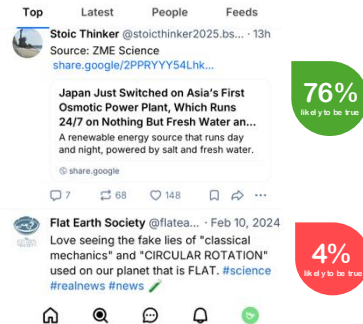
Contributors: Heather Lyons, Isabel Zuluaga, Edgar Garcia, Samantha Lee, Kalyani Shinde

Faculty Advisor: Professor Navid Shaghagh

Department: Computer Science & Engineering

## Overview

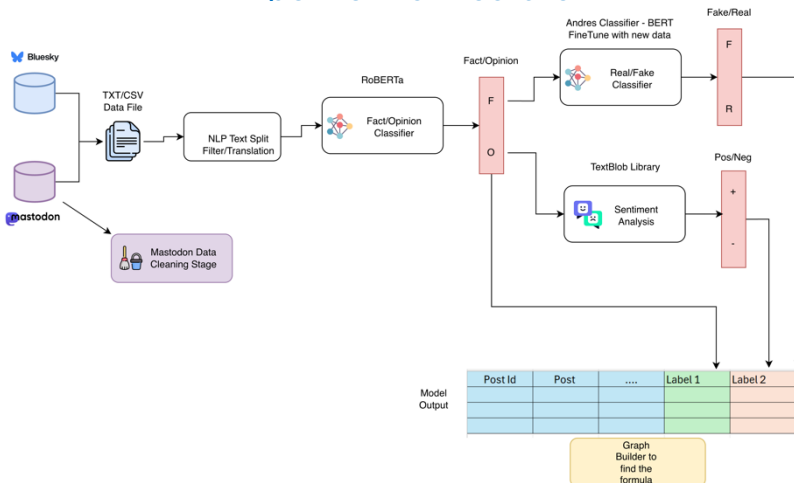
- The project's overall goal is to mitigate the spread of misinformation on social media sites.
- Background stats:
  - False news is 70% more likely to be retweeted than true news (Beauvais).
  - The top 1% of false news cascades routinely reaches 1000-100,000 people (Beauvais).
- The project aims to create a social media tool that shows users calculated initial credibility scores of both the accounts and the posts they see on their feed which are then dynamically updated through each interaction. In this way fake news posts will be buried far enough down the feed that most people will never see them.
- To accomplish this, a feed generator algorithm that receives posts from the BlueSky + AT Protocol firehose is developed that sends the posts through a pipeline of ML models, and outputs a credibility score.
- From there, the tool ranks the posts a user sees with high credibility scores at the top and low credibility scores at the bottom.



Source: Beauvais, Catherine. "Fake news: Why do we believe it?." *Joint Bone Spine* vol. 89,4 (2022): 105371. doi:10.1016/j.jbspin.2022.105371

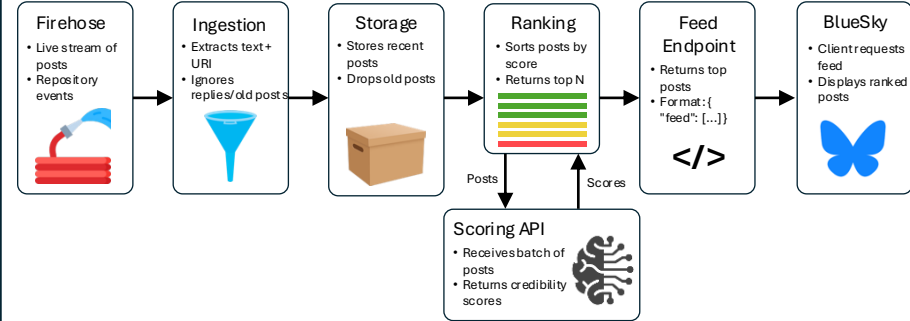
Example of what the tool could look like.

## Pipeline Architecture



Each post will pass through several ML models to determine its credibility score, if applicable.

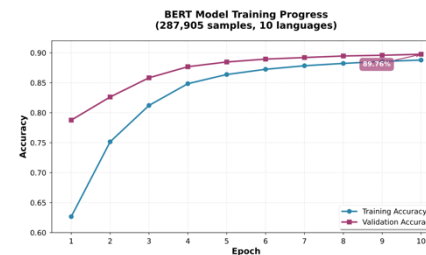
## Feed Generator Algorithm



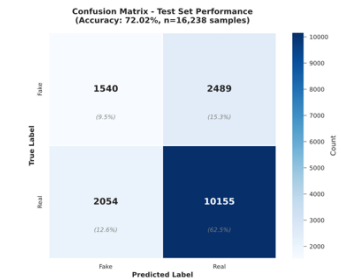
The BlueSky feed generator algorithm receives posts from the firehose, sends them to an API that assigns credibility scores, receives those results, and then ranks them on a user's feed.

## BERT Model

- Bidirectional Encoder Representation from Transformers (BERT) process text from left and right sides simultaneously, allowing it to learn contextual relations between words rather than just sequential order.
- The BERT model for this project is trained on ~300,000 samples in total.
- Its training data comes from 6 datasets that feature posts in different languages about topics like celebrity gossip, political news, COVID-19, and more.



Training and validation accuracy across 10 epochs.



Confusion matrix after testing ~16,000 samples.

## Next Steps

- Create an API that the feed generator can hit for the ML pipeline and thus make it available publicly.
- Add a credibility score for the user accounts that is continually updated based on their posts in order to better bury both the fake news posts in the post feeds as well as the fake news generating accounts themselves.